

- 6** 10 cm³ of a hydrocarbon was mixed with 50 cm³ of oxygen and combusted completely. After the resulting gas mixture was cooled and passed through aqueous sodium hydroxide, the volume of the residual gas mixture was found to be $\frac{1}{4}$ the volume of the original gas mixture before combustion. All volumes were measured at room temperature and pressure.

What is the formula of the hydrocarbon?

- A** C₂H₄
- B** C₂H₆
- C** C₃H₆
- D** C₃H₈